

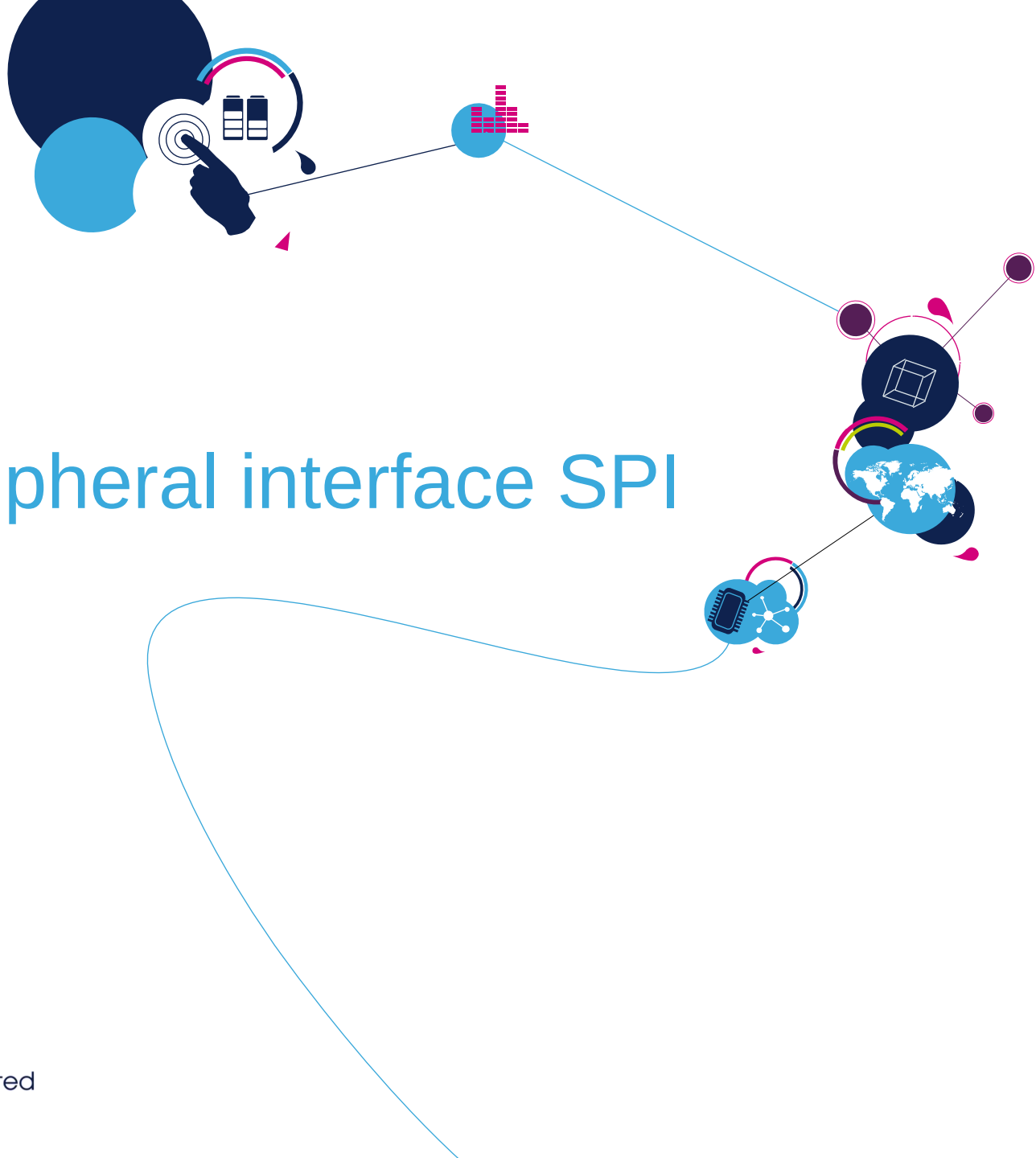


# STM32F3 Technical Training

**For reference only**

**Refer to the latest documents for details**





# Serial peripheral interface SPI

# SPI Features (1/2)

3

- Full duplex synchronous transfers (3 lines)
- Half duplex/Simplex synchronous transfers (2 lines, bi-directional data line at half duplex)
- Programmable clock polarity & phase, data MSB/LSB first
- Master/multi Master/Slave operation
- Dynamic software/hardware NSS management (Master/Slave)
- Hardware CRC feature (8-bit & 16-bit data frames checking)
- Flags with IT capability (TxE, RxNE, MODF, OVR, CRCERR)
- Programmable bit rate: **up to  $f_{PCLK}/2$**
- BSY flag (ongoing communication check)
- DMA capability (separated Rx/Tx channels, automatic CRC & Tx/Rx access/threshold handling)





- **New enhanced NSS control:**
  - NSS pulse mode (NSSP)
  - TI mode
- **Programmable data frame from 4-bit to 16-bit**
- **Two 32-bit Tx/Rx FIFO buffers with DMA capability**
- **Data packed mode control**

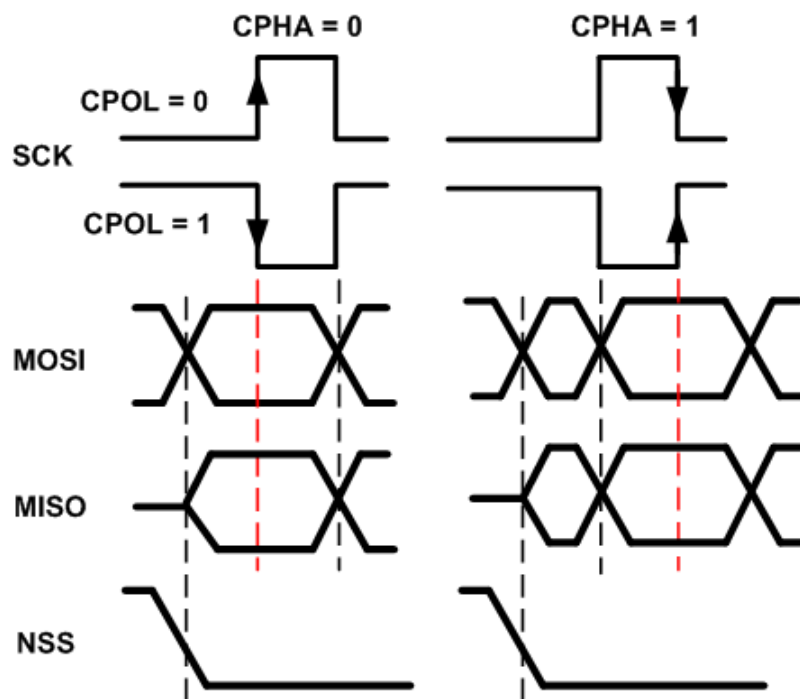
## 5



# Data Frame Format (Motorola mode)

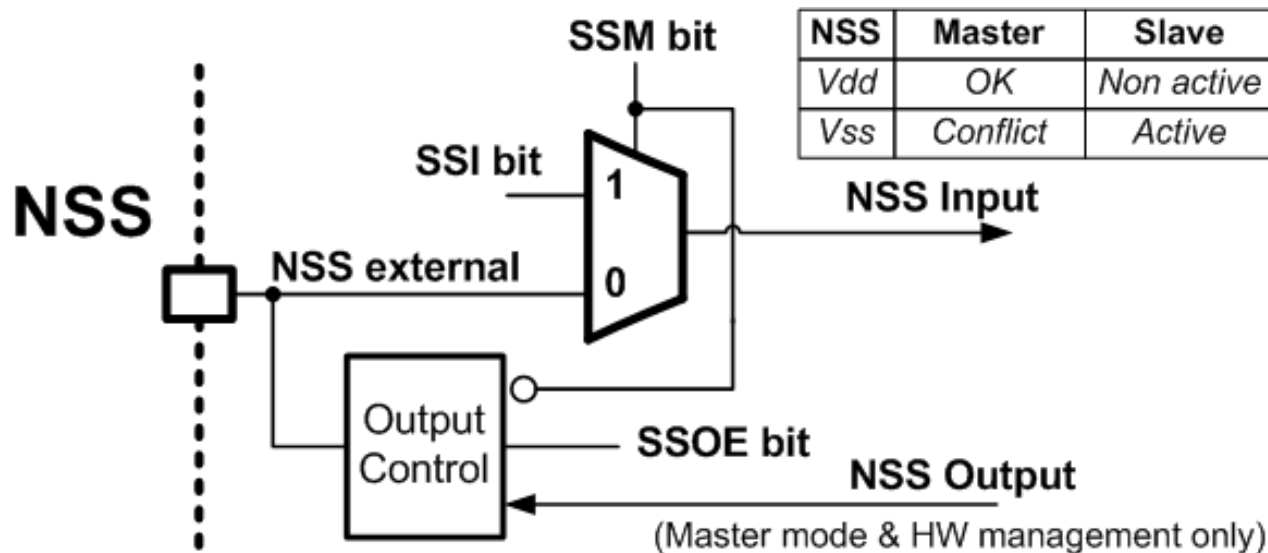
6

- Data frame format :
  - Programmable size of transfer data frame format from 4-bit up to 16-bit
  - Programmable clock polarity & phase, data bit order MSB/LSB first



# NSS management 7

- **NSS input** – SSM selects HW control (NSS pin) or SW control (SSI bit):
  - Slave mode - select slave for communication  
(optionally can be used for synchronization of a transaction begin)
  - Master mode - signalize conflict between masters
- **NSS output** – HW control at master mode only  
(SSM=0, SSOE=1 or at specific modes - TI, NSSP)

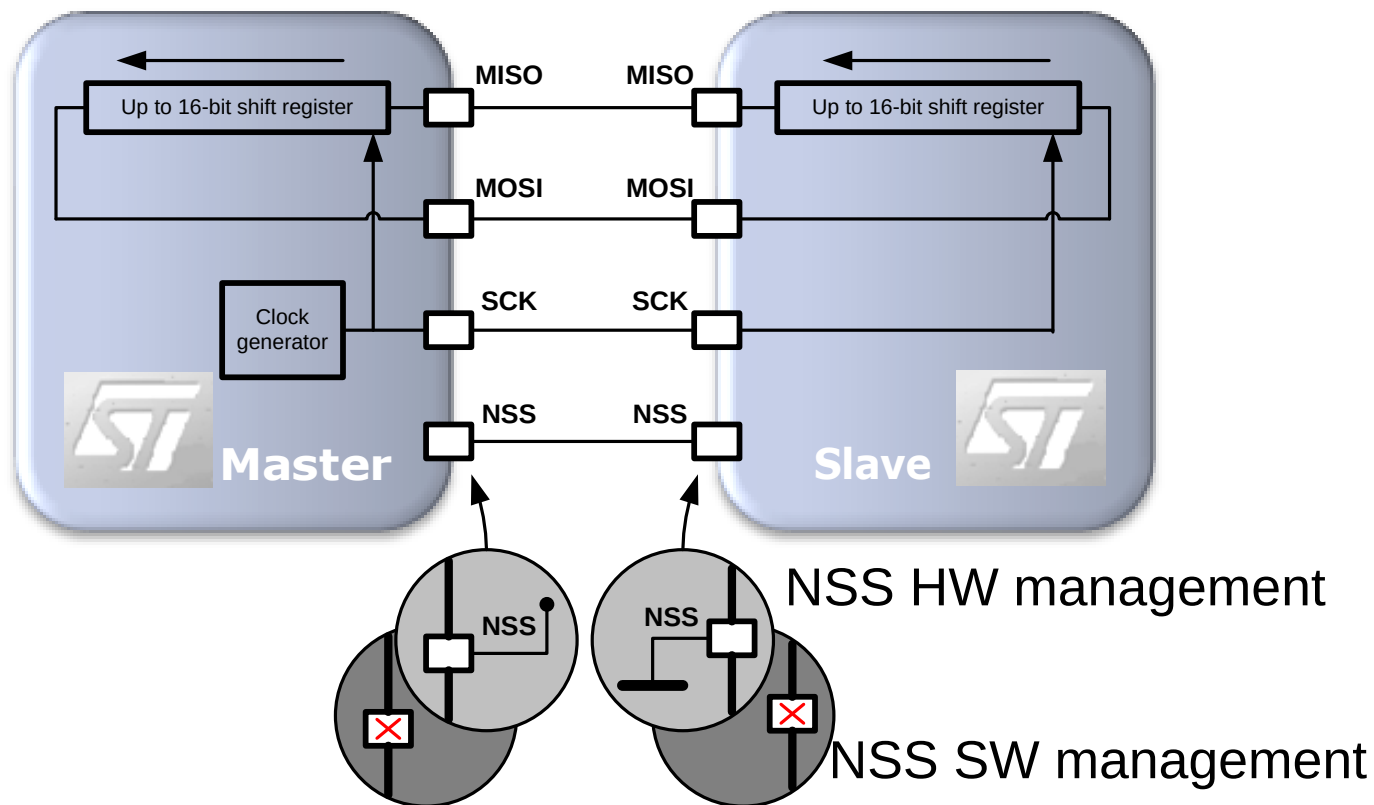


# Communication modes 1/3

8

- **Full Duplex mode** (Single master & single slave)

Three lines are necessary at least - MISO, MOSI, SCK (NSS is optional)

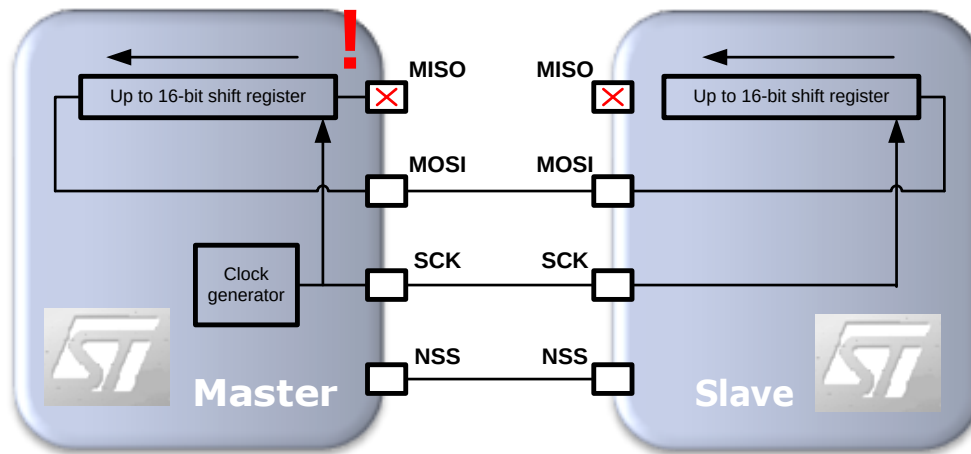




# Communication modes 2/3

9

- **Simplex mode** (Single master & single slave)
  - Two lines are necessary at least (NSS is optional):
  - MISO & SCK (Master Rx only, Slave Tx only)
  - MOSI & SCK (Master Tx only, Slave Rx only - at below figure)

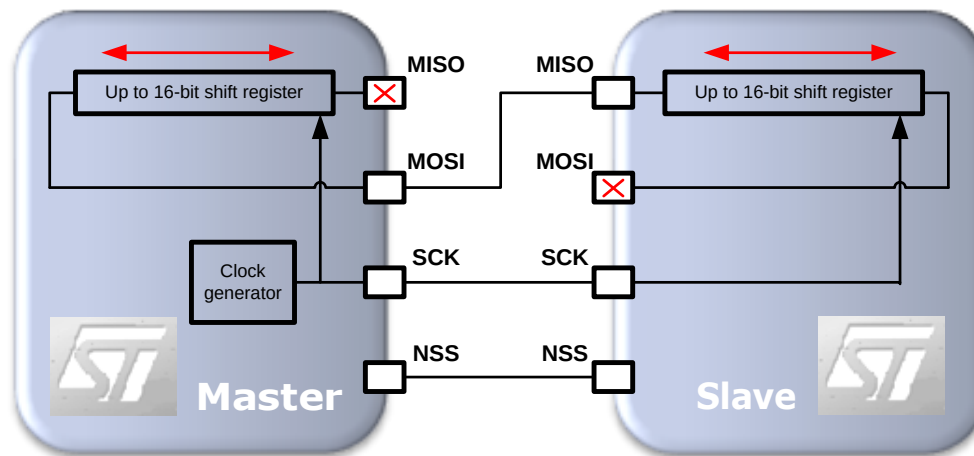


# Communication modes 3/3

10

- **Half duplex mode** (Single master & single slave)

Two lines are necessary at least - bidirectional cross data line MOSI-MISO and SCK (NSS is optional)



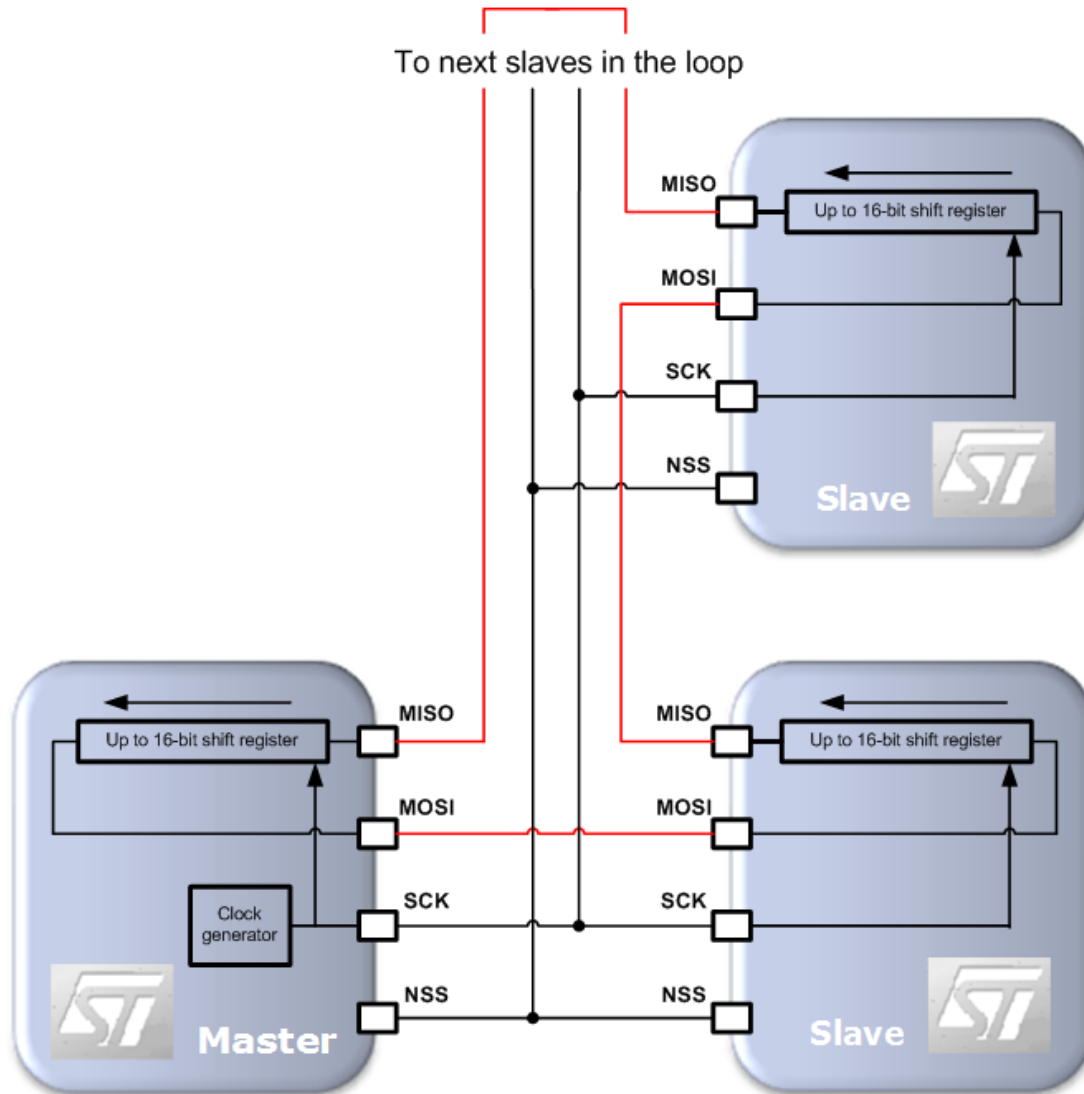
## 11



- 11

# Multi slave circular duplex chain

12



- Data lines are connected into a closed loop
- Master must shift the data through all of the slaves
- One common NSS is used

# 32-bit Rx and Tx FIFOs

13

- Two separated 32-bit FIFOs for receive and transmission
- 8-bit or 16-bit read/write access to FIFOs.
- Occupancy level flags FTLVL[1:0], FRLVL[1:0], TxNE, RxNE
- Different capability of Tx and Rx FIFOs

| Rx & Tx FIFO occupancy |       |       | TxNE | RxNE   |       |
|------------------------|-------|-------|------|--------|-------|
| 16-bit                 | 8-bit | FxLVL |      | 16-bit | 8-bit |
| 0                      |       | 00    | 1    | 0      | 0     |
| 1/4                    |       | 01    | 1    | 0      | 1     |
| 1/2                    |       | 10    | 1    | 1      | 1     |
| >1/2                   |       | 11    | 0    | 1      | 1     |

FIFO Access

FIFO Threshold

\*) Max 3x 8-bit for Tx FIFO, 4x 8-bit for Rx FIFO



- DMA handles automatically:
  - all the TXE and RXNE events
  - CRC is sent after last data frame
  - Initialization of CRC calculation
  - change of data register access and Rx threshold control of the last data frame in case it is an odd frame at packed mode  
(user must set *LDMA\_TX* & *LDMA\_RX* bits at this case!)
- Packed mode is used by DMA:
  - data frame fits into one byte
  - when peripheral size (PSIZE) is set to 16-bit for SPI DMA channel

### Notes:

- OVR flag is set at transmit only mode (SPI continues to receive)
- BSY & FTLVL must be checked before SPI is disabled

- Communication activity checking  
(to prevent corruption of ongoing transfer)
  - Before SPI or its clock are disabled (at some modes)
  - Before entry to Halt
- Cleared under any one of the following conditions:
  - When the SPI is properly disabled
  - When a master mode fault occurs (MODF=1)
  - In master mode between transactions when no next pattern is ready to transfer (FTLVL=00)
  - In slave mode between each data pattern transfer

*Note:* BSY & FTLVL must be checked before SPI is disabled



# CRC calculation (1/3)

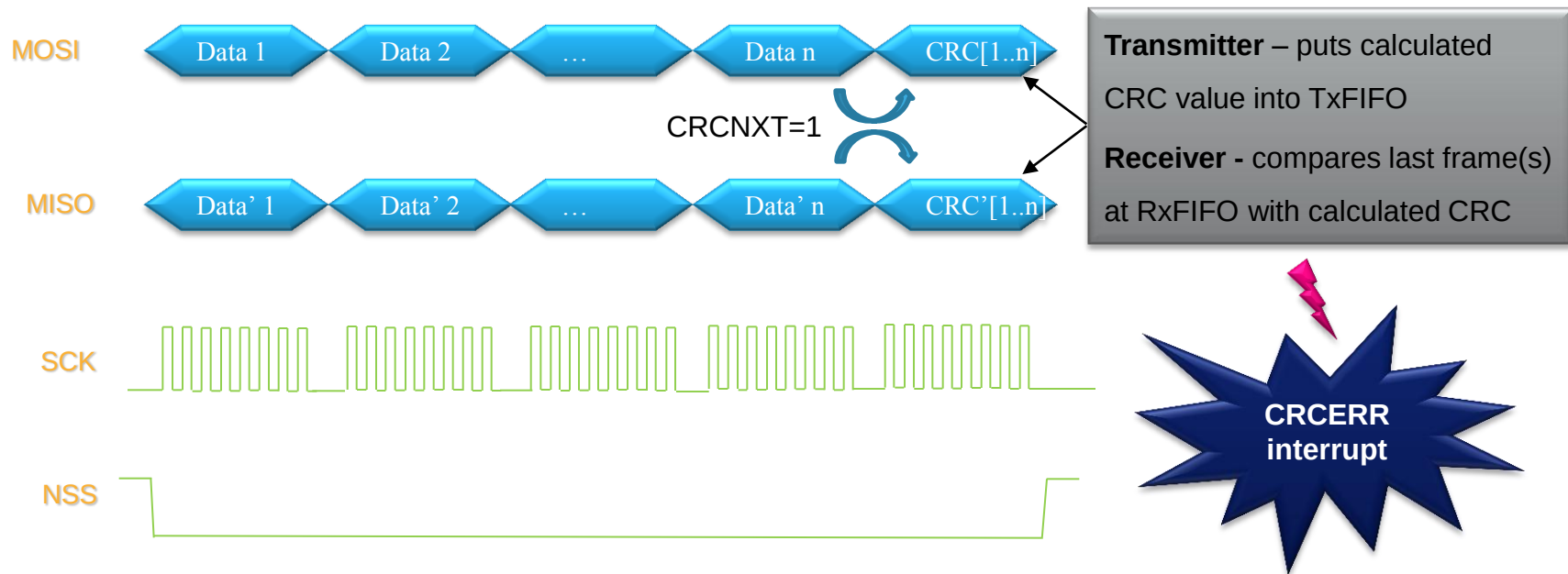
17

- Hardware CRC feature for reliable communication:
  - Separated CRC calculators implemented for transmitted and received data
  - CRC value is transmitted as a last transaction(s)
  - CRC error checking for last received byte and interrupt generation on error
  - Programmable CRC polynomial calculation (odd polynomials only)
  - Available for 8-bit or 16-bit data patterns only
  - Two possible CRC calculations: CRC8, CRC16-CCITT standard

# CRC calculation (2/3)

18

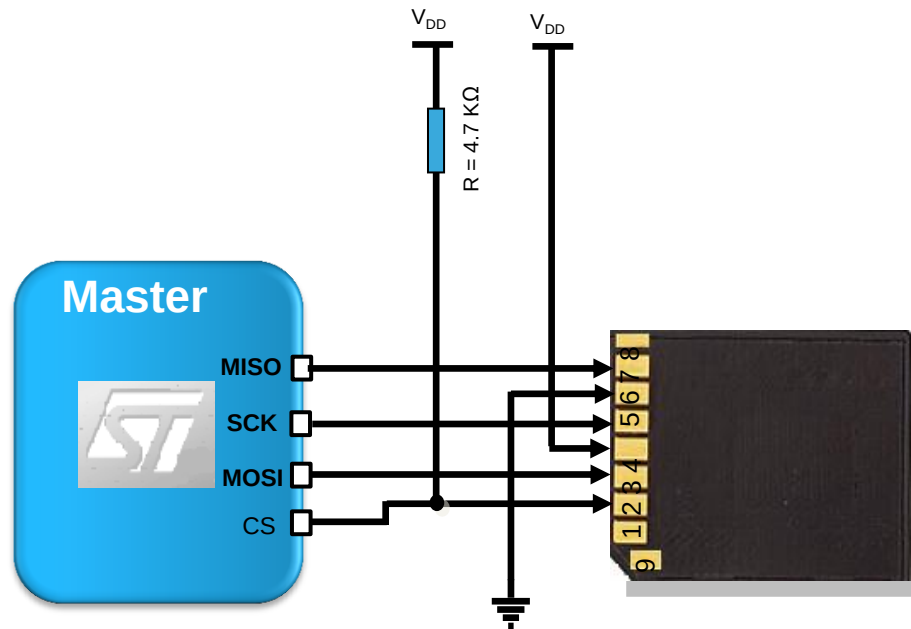
- Example of n data transfer between two SPIs followed by the CRC transmission of each one in Full-duplex mode
- SPI\_TXCRCR and SPI\_RXCRCR – separated registers for CRC calculation



# CRC calculation (3/3)

19

- Basic SD/MMC support (SPI protocol):
  - Performance: speed up to 18MHz
  - Error checking: hardware CRC calculation

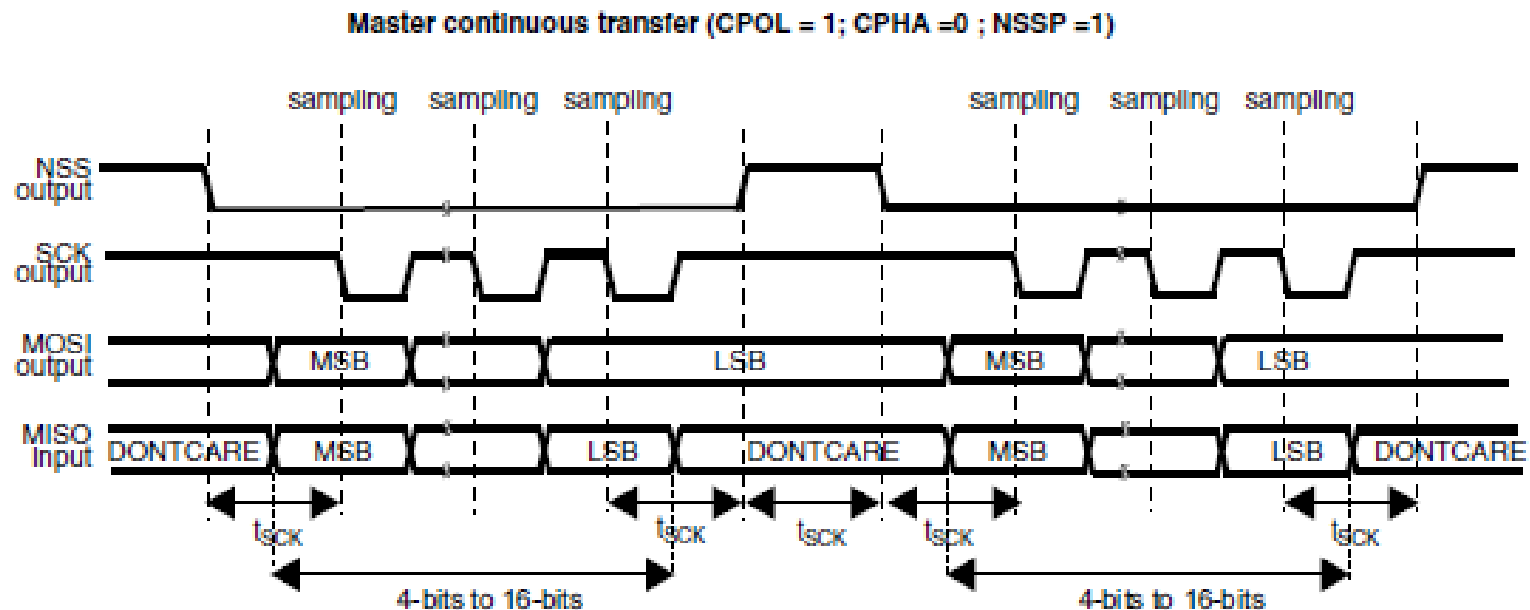


# NSS enhanced modes (1/2)

20

## Pulse mode (NSSP=1)

- At master and Motorola mode with CPHA=0 only
- NSS output is managed by HW

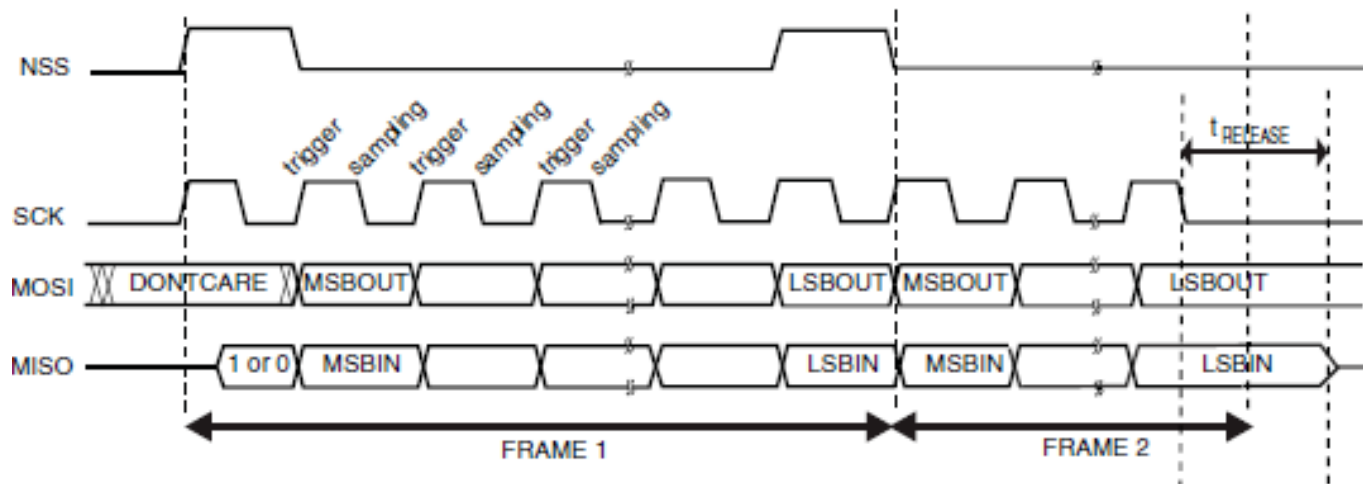


# NSS enhanced modes (2/2)

21

## TI mode (FRF=1)

- Clock and NSS are managed by HW
- At slave, baud rate setting defines t<sub>RELEASE</sub> (MISOs HiZ)
- Format frame error interrupt (FRE)



## When data packed mode is used

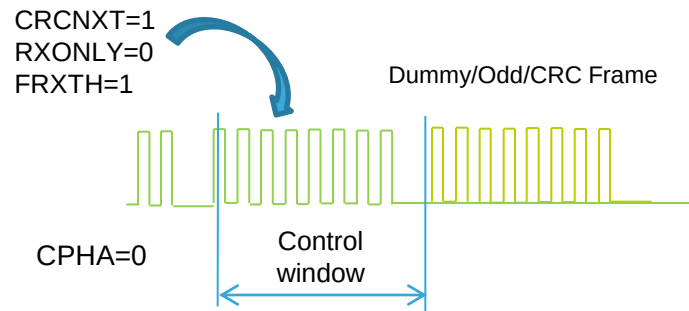
- Keep Rx threshold & read access of Rx FIFO always in line either 8-bit or 16-bit (preferable -> limited number of events)
- Change Rx threshold just before last odd data frame is received

## When go to Halt or when disable the SPI

- check read FIFO occupancy and bus activity ( $FxLVL[1:0] = 00$  &  $BSY = 0$ )

## When communication is continuous (e.g. master Rx-only)

- Perform Rx threshold, change/CRC control or Stop within “Control window”



- What is a maximum SPI speed?
- When data packed mode is used?
- What should be in line when access the RxFIFO?
- Does a master take any care of the NSS signal level if it is managed by SW?
- How many bytes (maximum) can be stored into TxFIFO till it is full when 8-bit access is used?
- What should precede before SPI is disabled?



Thank you !



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